Personalization of Speech Therapy Treatment with Socially Assistive Robots: Implications for Aphasia Rehabilitation

Shifa Somji[1], Ji Yeon Lee[2] and Sooyeon Jeong[1]

Abstract— **Aphasia is a post-stroke condition that results in many communication deficits and negatively impacts an individual's ability in language production and comprehension, leading to limited participation in daily activities. People with aphasia (PWA) often experience barriers to appropriate care due to limited health coverage and clinician availability. In our previous work, we investigated language alignment behavior to show that socially assistive robots (SARs) can provide scalable speech therapy. This study explores opportunities for using large language models (LLMs) to personalize the speech therapy treatment and determine the efficacy of personalization. We developed three autonomous robot interaction behaviors and compared them in a between-subject experimental study: (1) a *control behavior* that was manually programmed and provided generic responses to PWA, (2) a *personalization behavior*, in which the SAR produced unique real-time responses for each PWA, and (3) an *personalization + interaction style behavior*, in which the robot not only produced unique responses, but also changed its interaction style based on the PWA's responses. Twenty-five PWA were randomized into one of the three experimental conditions and interacted with a Furhat and NAO robot in an interactive language task and simple speech therapy exercises. Results from our study show that SARs can deliver successful personalized speech language therapy and adapt its interaction style based on each PWA's unique needs and preferences in a home context. Our work provides meaningful insights into how SARs can be integrated into PWA's homes to effectively provide long-term language retention and aphasia rehabilitation.**

I. INTRODUCTION

Aphasia is a language disorder that affects one's ability to use or understand language [1]. Every year about 180 thousand people are newly diagnosed with aphasia in the United States [1]. Aphasia can affect people of all ages but is mostly common in older adults (OA) – people over the age of 65 – and primarily occurs due to a stroke [2]. It is provoked by dysfunction in specific brain regions, usually in the left cerebral hemisphere [3]. The main symptoms of aphasia include difficulty with language comprehension, spontaneous speech, and repetition abilities [3]. People with aphasia (PWA) have trouble forming their thoughts into sentences that are both grammatically correct and meaningful [2]. As a result, PWA experience significant difficulty in sustaining independent daily lives and interacting with others, including family and friends.

Among several treatment methods for aphasia, speech- language therapy (SLT) is the most common and effective treatment for aphasia rehabilitation. SLT aims to either decrease language impairment or eliminate communication barriers by targeting subcomponents of language, such as phonology and syntax, or improving sentence delivery [4]. SLT was shown to positively impact PWA's language comprehension and production [4]. However, many PWA experience barriers to sustainable speech therapy services, such as limited insurance coverage, geographical distance, and a shortage of speech-language pathologists [5]. Previous findings suggest that PWA need frequent and intensive SLT for therapeutic outcomes [6] but many PWA are only eligible to receive few (10-20 sessions) therapy sessions per year due to limitations in healthcare coverage and lack of available trained speech-language therapists [7]. We aim to bridge this gap by developing a SAR that can deliver scalable and easily accessible SLT for PWA in a home setting.

Previous studies found that SARs can successfully support language learning tasks [8]–[10], especially for young children. Spitale et al. [11] found that interacting with either physical or virtual SARs resulted in significant linguistic improvement for children with language impairment. However, most existing work focuses on alignment on the individual word level and little work has explored structural alignment effects in human-robot interactions, i.e., mirroring the sentence

[1]Shifa Somji and Sooyeon Jeong are affiliated with the Department of Computer Science at Purdue University. ssomji@purdue.edu

[2]Ji Yeon Lee is affiliated with the Department of Speech, Language, and Hearing Services at Purdue University. lee1704@purdue.edu

structure used by the other interlocutor. In addition, no work has been done to investigate the therapeutic efficacy of SARs in supporting aphasia patients. Our previous work explored the effect of implicit structural alignment, which has been shown to facilitate sentence processing in PWA, in contrast to explicit instructions, which is not always effective for PWA [12].

In this work, we develop an SAR that provides personalized SLT intervention based on each PWA's unique needs and linguistic skill level, and tailor its interaction style (e.g., perceived role) for their preference in long-term in-home interactions. We address the following research questions:

R1. How can we develop a SAR that provides scalable and adaptive SLT for PWA's language rehabilitation?

R2. How can we personalize and adapt the SAR's SLT content and interaction style for each PWA for long-term engagement and therapeutic outcomes?

II. BACKGROUND

A. Speech-Language Therapy for Aphasia

Previous research has shown that SLT needs to be conducted as often and as intensely as possible in order to be effective [1]. However, many PWA experience barriers to sustainable speech therapy services, such as limited insurance coverage, geographical distance, and a shortage of speech-language pathologists [2]. In the current healthcare system, receiving short-term high-intensive therapy by speech-language therapists is difficult due to limited funding by health insurers and the already existing large caseload of aphasia therapists [3]. Thus, there exists a need for scalable and affordable interventions that can support patients with aphasia in a long-term home context.

B. Interactive Technologies for Speech-Language Therapy

In order to accommodate some of the challenges with existing SLT, other technologies have been proposed to extend SLT, including tele-health therapy [4] and mobile-based interventions [5], [6]. Tele-health therapy eliminates travel time and provides timely and family-centric rehabilitation care for PWA but still requires a human therapist that again may not be funded by health insurers. Mobile-based interventions can increase ease of access to care at a distance and provide personalized support. However, several studies have found that older adults often experience barriers to using mobile health (mHealth) technologies, which can potentially hinder their motivation to benefit from mHealth intervention [7].

In comparison to other interactive technologies, SARs can be a solution to improve access to personalized care, training, and rehabilitation for large populations to enhance their quality of life [8]. SARs were shown to be effective as a therapeutic tool for children [9], the elderly [10], stroke patients [11], and other special-needs populations [12] that require personalized care. Previous studies also showed that robots can be used to enhance people's psychological and physiological well-being and can be used in the rehabilitation and treatment for many disorders [13], [14], and we believe SARs can also be used to extend SLT for scalable aphasia treatment.

III. EXPERIMENTAL STUDY

A. Participants

Twenty five PWA (14 women, 11 men) were enrolled in this study. PWA were between 35 to 82 years old (M = 61.4, SD = 15.9) and completed between 12 to 19 years of education (M = 15.4, SD = 2.5). Fifteen age- and education-matched control participants were also enrolled in this study. Participants were recruited from the Greater Lafayette area, through social media, or referred from aphasia programs in other areas. Our study protocol was reviewed and approved by the Institutional Review Board, and all participants signed the consent form prior to the study.

All participants were native speakers of American English with at least an 8th-grade education and reported no history of neurological or psychological disorders. PWA had no history of neurological or psychological disorders prior to their stroke. All participants self-reported that they had normal or corrected-to-normal hearing and passed an informal hearing screening (500, 1000, and 2000 Hz pure tones at 40 dB at 80% computer volume without headphones). They also self-reported having normal or corrected-to-normal vision and passed an informal near vision screening at a distance of 16 inches using a digital version of the LEA near vision screener (adapted from Hyvärinen, 2018). These screenings were used to ensure that participants could hear and see the stimuli that were used in the task. In addition, control participants completed the Mini Mental State Examination (MMSE; Folstein, Folstein, & McHugh, 1975) and scored within normal limits for cognition based on their age and education (M = 29.1, SD = 1.5).

B. Language Testing for PWA

Each PWA had a diagnosis of aphasia with a post-stroke onset time of between 24 and 177 months (M = 67.1, SD = 54.2), as verified by their medical records. PWA were assessed using a combination of language tests. Given that the experimental task involved sentence-level stimuli, PWA with mild-to-moderate aphasia severity were recruited as indicated by the Aphasia Quotient (AQ) from the Western Aphasia Battery-Revised (WAB-R; Kertesz, 2006). All PWA showed relatively preserved auditory comprehension of single words and different types of simple sentences on the Auditory Comprehension test of the WAB-R. Their comprehension of verbs was also confirmed to be relatively intact, using the Verb Comprehension Test of the Northwestern Assessment of Verbs and Sentences (NAVS; Cho-Reyes & Thompson, 2012). PWA were able to comprehend and produce simple sentences but were impaired when comprehending and producing more complex sentences as shown in their performance on the Argument Structure Production Test and Sentence Comprehension Test of the NAVS. Participants who completed the study remotely were assessed using remote-adapted versions of the WAB-R and NAVS following the same procedures as previous studies (Dekhtyar et al., 2020; Keen et al., 2021; Pearson Assessments, 2021).

C. Experimental Design

We designed three different robot behaviors for our study: the control condition (*control*), the personalization only condition (*PS*), and the personalization & interaction Listening condition (*PS + IN*).

In the *control* condition, the Furhat and NAO robots only asked and replied based on the pre-scripted interaction flow, but did not exhibit any unique behaviors during the interaction. In the *PS* condition, the SARs used GPT-4o to produce unique responses to the PWA's answers. Lastly, in the *PS + IN* condition, the SARs not only used GPT-4o again, but also included facial expressions and changed their interaction style based on the PWA's responses.

The design architecture consists of two core components: Sentiment-based Prompting and Active Listening. After receiving raw audio of human speech, OpenSmile [15] produces audio features. The Automatic Speech Recognition (ASR) of the Whisper model[1] recognizes speech in real-time. Based on audio features from OpenSmile and detected sentiment labels from the GPT-4o-based sentiment analyzer, we generate facial expressions and appropriate body movements during the SAR's interaction. We also appropriately evaluate the sentiment of the speech. We used an intricately crafted prompt for GPT-4o [16] to classify the sentiment of the verbal utterance produced by the study participants. We assign -1 for negative sentiment, 0 for neutral sentiment, and 1 for positive sentiment. The sentiment was then used in the prompting process to modify the SAR's response by mimicking the PWA's tone.

[1]https://learn.microsoft.com/en-us/azure/ai-services/speech-service/whisper-overview

D. Experimental Tasks

All participants completed the experimental sequence which consisted of a pre-test, a priming block, and a post-test in a single experimental session. As shown in Figure 1, the pre-test consisted of a sentence-picture matching task where participants heard and saw a target sentence with an ambiguous prepositional phrase presented along with a pair of pictures that allowed for either a low attachment (LA) or high attachment (HA) interpretation. All participants in both groups chose HA pictures more than 80% of the time in the pre-test, so they were considered to show a preference for HA interpretations of the target stimuli.

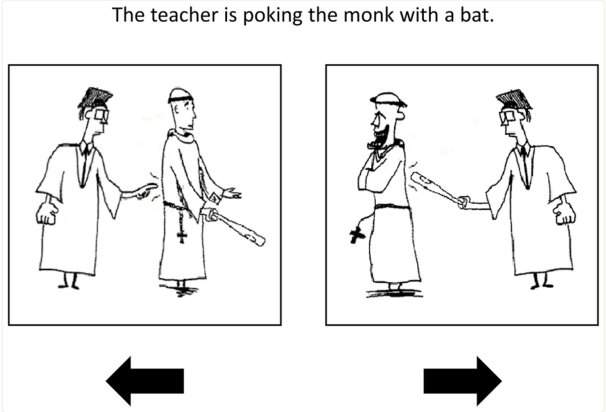

Fig. 1: Example of a comprehension target trial in the pre- and post-tests. The black left and right arrows on the screen corresponded to the left and right arrow keys on the participant's keyboard which were used to select pictures.

After the pre-test, participants completed a priming block that contained the production-to-comprehension priming task. The priming block consisted of prime-target pairs, which were used to test whether priming the opposite interpretation from the participant's initial preference in the pre-test would significantly influence

them to adopt the primed interpretation. Since participants showed a HA preference (selected more than 80% HA pictures in the pre-test), they were all trained on the LA priming block (100% LA primes). The rationale was that in order to induce a larger priming effect, using the participant's less preferred structure should produce more prediction error and therefore increase priming as a result of the inverse frequency effect (Hartsuiker & Westenberg, 2000; Kaschak, Kutta, & Jones, 2011; Scheepers, 2003). After a brief break, participants started the post-test. The post-test consisted of the same sentence-picture matching task as the pre-test. However, the post-test included different sentence and picture stimuli from the pre-test and the priming block to ensure that participants' performance in the pre-test was not due to repetition effects of the stimuli.

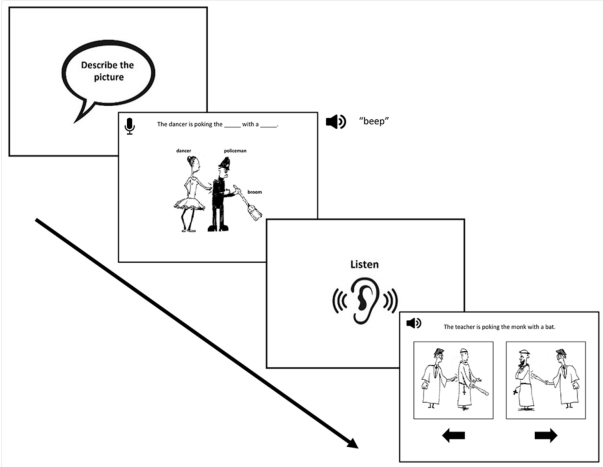

Fig. 2: Example of a production prime-comprehension target trial in the priming block.

E. Priming Block

Figure 2 shows an example of a production prime-comprehension target pair in the priming block. For the production of prime sentences, a cued sentence completion task was employed, similar to Branigan et al. (2005). A prime sentence template with blanks for the goal and theme was presented at the top of participants' screen to ensure that PWA were able to produce the prime sentence in the required LA structure. Participants were instructed that after they saw *"Describe the picture"* on the screen, they would hear a beep sound and see a picture. Participants described the picture by using the sentence at the top of the screen and filling in the blanks of the sentence using all the labeled words in the picture (e.g., The dancer is poking the [policeman] with a [broom]). Importantly, different from comprehension target items, the prime items were accompanied by only one picture which allowed for a LA interpretation, ensuring that participants were producing LA sentences.

Participants completed a set of 6 prime-target trials as practice, and then participants began the priming block. Immediate feedback on the production accuracy of the prime sentences was given for the practice trials, but only neutral feedback (e.g., *"you're doing fine"*) was provided for experimental trials.

F. Post-Study Interviews

After the study ended, participants were asked to provide qualitative feedback on their experience interacting with each SAR via a semi-structured interview. Through this interview, we aim to understand PWA's interaction experience with Furhat and NAO in comparison to the interaction with a human partner and identify key similarities and differences in these experiences. The questions in the survey aim to test participants' comfort and ease in interacting with each partner.

We also conducted a semi-structured interview with PWA caregivers. During this interview, we asked them questions on how SARs can supplement therapy sessions at home and can be useful to PWA beyond speech-language therapy.

IV. DATA ANALYSIS

A. Video and Audio Data

All interactions were video/audio recorded and transcribed with Whisper transcription and Microsoft Word transcription and the research team manually checked the AI-generated transcriptions for any errors. The camera was placed in front of the participants to fully capture the facial expressions of the participants. The session was recorded when the participants began to fill out the pre-questionaire and finished when the participants left the study room. The data is then uploaded to the REED folder on the University cluster. The audio recordings were manually transcribed.

V. RESULTS

A repeated-measure ANOVA test revealed that the type of interaction behavior did have a statistically significant effect on participants' production accuracy for prime sentences ($F(2, 27)$=0.93, p=0.03). Specifically, post-hoc Dunn's tests demonstrated statistical significance in *PS + IN* and *control* (p=0.002); *PS*

+ *control* (p = 0.003), but there were no significant differences for *PS* + *IN* and *PS* (p = 0.41).

Paired t-tests showed that there is no statistically significant difference in how participants perceive the usefulness (*NAO* M=3.69, SD=0.97; *Furhat* M=3.63, SD=1.04; $t(11)$=0.22, p=0.83), ease of use (NAO M=2.86, SD=1.36; *Furhat* M=3.22, SD=1.15; $t(11)$=-1.65, p=0.13), or appearance (*NAO* M=3.73, SD=1.20; *Furhat* M=3.79, SD=1.03; $t(11)$=-0.51, p=0.62) between the two robots.

From the semi-structured interview with participants, some participants enjoyed NAO's physical body (P7, P8, P9, P11), while other participants thought NAO's movements were too distracting (P1, P4, P6, P9). Sixteen participants noted that Furhat was very human-like (P2, P4, P5, P6, P8, P10, P11, P13, P14, P16, P17, P18, P20, P22, P24, P25), although sometimes realistic in an unnatural way (P3, P8, P10, P11, P15, P21, P23). P3 thought that "NAO was friendlier, while Furhat was stern, but easier to understand" and Furhat reminded him of "the giant head projections at Disney World, making it more realistic...looking at me very often." P14 shared that "Furhat had a more realistic voice, with its cadence, tone, stress" while "NAO was wonky and mechanical...feeling like a lego set for a young child." P21 described that "when it comes to robotic things, you want something fancy, something different from humans, so I would want something like the first robot [NAO] in my home." P6 thought that "NAO had features that were more sci-fi and fictitious, but Furhat was very realistic...both robots were very clear and easy to understand." Overall, patients thought NAO was cartoon-like and Furhat would be more useful in a long-term context.

From the semi-structured interview with caregivers, we received very positive feedback. Many caregivers noted that the "robots could provide 24/7 support for PWA" (caregivers for P3, P4, P7, P9, P11, P14, P18, P20). Some suggested that "language rehabilitation would be highly successful with more frequent communication with both a robot and speech therapist" (caregivers for P2, P5, P6, P12, P19, P23, P24). Additionally, caregivers agreed that "robots could help PWA feel less alone...especially with a communication partner in their homes" (caregivers for P3, P5, P13, P20, P25). Overall, caregivers saw the long-term potential for using robots to provide personalizable and frequent SLT in homes of PWA.

VI. DISCUSSION

Our findings demonstrate that socially assistive robots (SARs) can successfully deliver personalized speech-language therapy and adapt their interaction styles to meet the unique needs of people with aphasia (PWA). By leveraging large language models (LLMs), SARs were able to generate real-time personalized responses, enhancing engagement and interaction quality. These results support previous research indicating that SARs can effectively promote communication and social engagement [17] and perform comparably to human interlocutors in task-based interactions [18].

Furthermore, our study highlights the feasibility of integrating SARs into home-based therapy for PWA. While participants were generally receptive to interacting with both Furhat and NAO robots, qualitative interview data revealed nuanced differences in their experiences, suggesting that factors such as robot design, voice, and response style may influence engagement.

The main limitation of our work is that we did not implement an advanced form of end-of-speech detection. Several participants took more time to think for their answers and our naive end-of-speech detection system failed to detect end-of-user response. Our future work will focus on investigating how to correctly detect the end-of-speech of the users.

VII. CONCLUSION

This study demonstrates the potential of socially assistive robots (SARs) to deliver personalized and adaptive speech therapy for people with aphasia (PWA). By leveraging large language models (LLMs), we developed autonomous robot behaviors that provided real-time personalized responses and dynamically adjusted interaction styles based on user input. Our findings indicate that SARs can effectively support speech-language therapy by adapting to individual needs and preferences, fostering a more engaging and supportive rehabilitation experience.

The ability of SARs to operate in a home setting provides a scalable and accessible solution for PWA, addressing barriers related to clinician availability and limited healthcare coverage. Future work should explore the long-term impact of SAR-driven therapy on language retention and functional communication outcomes. Additionally, further refinement of personalization techniques and multimodal interaction strategies could enhance the effectiveness of SARs in aphasia rehabilitation. By integrating SARs into daily life, we can create sustainable, personalized interventions that

improve communication abilities and overall quality of life for individuals with aphasia.

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